Chapter 23 | Complete worked solutions

Object Detection and Segmentation

All eleven exercises are solved here. Trace boxes and mask pixels, distinguish training assignment from ranked evaluation, and design prompting and coordinate checks. The examples run on CPU without a model or dataset download. Three plots show mask overlap, precision envelopes and detection errors. The written PDF preserves all explanations and plots; the notebook PDF adds code and outputs. These are teaching fixtures, not benchmark results for a trained detector.

1. Compare two boxes

The box $A = (1, 1, 4, 3)$ has width $4 - 1 = 3$ and height $3 - 1 = 2$, hence area 6. The box $B = (2, 0, 5, 2)$ also has width 3 and height 2, hence area 6. Coordinates mark edges; no extra unit is added at either endpoint.

The intersection starts at the larger left edge and ends at the smaller right edge, giving horizontal interval $[2, 4]$ of width 2. Vertically it runs from 1 to 2, giving height 1. Its area is 2. Adding the two box areas counts the overlap twice, so subtract it once: union $= 6 + 6 - 2 = 10$ and IoU $= 2/10 = 1/5$.

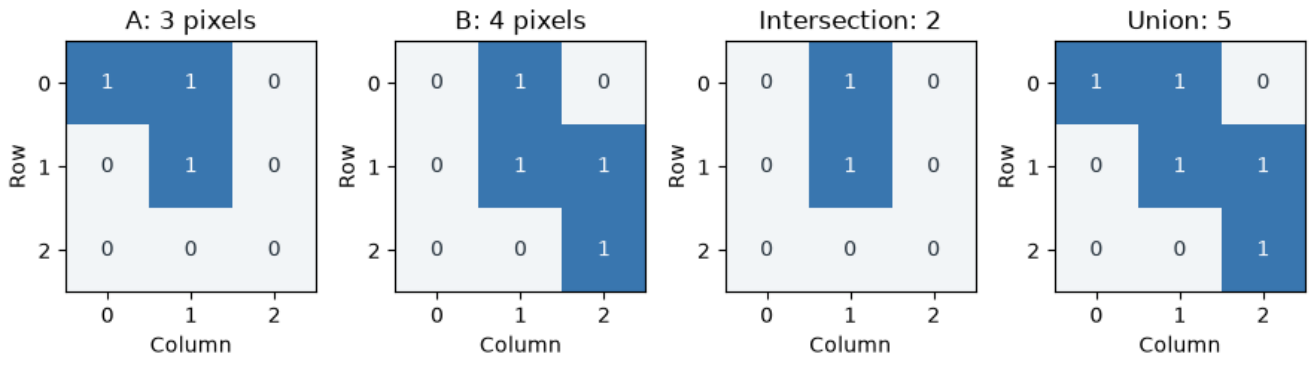
Moving B right by one changes both x edges: $B' = (3, 0, 6, 2)$. Its area stays 6. The overlap now has width 1 and height 1, hence area 1. The union is $6 + 6 - 1 = 11$, giving IoU $1/11$. Translation changed the intersection, not the individual box areas.

2. Count shared pixels

Use zero-based (row, column) coordinates. Rows increase downward and columns rightward; this order differs from an (x, y) point. The two masks are

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

The shared pixels are $(0, 1)$ and $(1, 1)$, so the intersection contains two pixels. The union contains $(0, 0)$, $(0, 1)$, $(1, 1)$, $(1, 2)$ and $(2, 2)$, five pixels in total. Equivalently, $|A \cup B| = 3 + 4 - 2 = 5$. Their IoU is $2/5$. The plot displays each mask and both set operations. A dark cell is foreground; the number inside each cell makes the count explicit.



3. Make a one-to-one assignment

The rows are Q1, Q2, Q3 and the columns are objects A, B. Choosing the minimum independently in each column would select Q1 twice, with apparent cost $2 + 1 = 3$. That is invalid: one query cannot stand for both objects. There are $3 \times 2 = 6$ valid assignments. Choose a query for A, then a different query for B:

Query for A	Query for B	Total cost
Q1	Q2	$2 + 5 = 7$
Q1	Q3	$2 + 2 = 4$
Q2	Q1	$3 + 1 = 4$
Q2	Q3	$3 + 2 = 5$
Q3	Q1	$4 + 1 = 5$
Q3	Q2	$4 + 5 = 9$

There are **two minima**, both of cost 4. Q1 for A and Q3 for B leaves Q2 unmatched. Q2 for A and Q1 for B leaves Q3 unmatched. Either is a valid optimal one-to-one assignment. The selected unmatched query receives the no-object class in training. One-to-one describes the constraint; it does not assert that the optimum is unique. Exercise 7 examines the helper's deterministic tie rule.

4. Track precision and recall

There are three annotated objects. A true positive increases the matched count; a false positive increases only the number of predictions. At each rank, divide matches so far by predictions seen for precision, and by three for recall:

Rank	Outcome	Matches	Precision	Recall
1	False positive	0	0	0
2	True positive	1	$1/2$	$1/3$
3	True positive	2	$2/3$	$2/3$
4	False positive	2	$1/2$	$2/3$

The maximum recall is $2/3$. Keeping all four predictions still leaves one object unmatched. Full recall requires a new qualifying detection, not merely a lower confidence cutoff on this unchanged list.

Scanning precision backward gives the envelope values $(2/3, 2/3, 2/3, 1/2)$. Recall increases are $(0, 1/3, 1/3, 0)$, so

$$\mathrm{AP} = 0(2/3) + (1/3)(2/3) + (1/3)(2/3) + 0(1/2) = 4/9.$$

The remaining recall interval from $2/3$ to 1 contributes zero. Dividing the area by the attained recall would instead give $2/3$, which is not this AP. The last false positive lowers final precision but adds no new recall width; removing it leaves the interpolated AP unchanged.

5. Check touching edges

Place box A at $(0, 0, 2, 3)$ and box B at $(2, 0, 4, 3)$. They share the vertical edge at $x = 2$. Move B a quarter unit left to $(1.75, 0, 3.75, 3)$, and the shared region becomes a thin rectangle.

The coordinates (x_0, y_0, x_1, y_1) mark continuous edges. The width is $x_1 - x_0$, and the height is $y_1 - y_0$. Both boxes have area $2 \cdot 3 = 6$ before and after the shift. The intersection width comes from the smaller right edge minus the larger left edge, clamped at zero. The intersection height is 3 in both cases.

When the edges touch, the overlap width is $2 - 2 = 0$. The common edge has zero area, so the union is $6 + 6 = 12$ and IoU is $0/12 = 0$.

After the shift, the overlap width is $2 - 1.75 = 0.25$. The intersection area is $0.25 \cdot 3 = 0.75$, and the union area is $6 + 6 - 0.75 = 11.25$. The IoU is $0.75/11.25 = 1/15 \approx 0.066667$.

The shift leaves both widths at two units and creates a positive overlap of exactly a quarter unit in width. The continuous-edge convention treats each endpoint as a boundary, so the width formula has no extra unit for an endpoint.

6. Inspect an empty union (optional)

Start with two 3×3 Boolean masks on the same pixel grid. Every entry is `False`, so neither mask marks any foreground. Add a foreground pixel at row 1, column 1 of the second mask.

The intersection counts pixels marked by both masks. The union counts pixels marked by either mask, counting a shared pixel once.

With both masks empty, the intersection and union both contain zero pixels. The ratio $0/0$ is undefined, and `mask_iou` returns `None`.

With one marked pixel in the second mask, the intersection is still empty and the union contains that one pixel. The ratio $0/1 = 0$ is a defined score for zero shared foreground.

The first pair has no foreground over which to measure overlap. The second pair does have foreground to compare, and the empty first mask shares none of it. The two return values preserve that difference.

7. Inspect matching indices

Use the three query rows from Exercise 3. The rows are Q1, Q2, and Q3, in that order. The columns are targets A and B. Both column minima pick Q1, for costs 2 and 1, so using those two minima would assign the same query twice.
Try each pair of distinct queries. The total for Q1 with A and Q3 with B, for example, is $2 + 2 = 4$. Increasing entry `[0, 1]` from 1 to 5 changes only the totals that pair Q1 with B.

Query for A	Query for B	Original total	Revised total
Q1	Q2	7	7
Q1	Q3	4	4
Q2	Q1	4	8
Q2	Q3	5	5
Q3	Q1	5	9
Q3	Q2	9	9

The original matrix has two equally good assignments. The helper tries query indices in ascending order for A, then for B. It keeps the first minimum, Q1 for A and Q3 for B, and reports zero-based `(query, target)` pairs, so this is `((0, 0), (2, 1))` and query index 1 (Q2) is unmatched. The other minimum, Q2 for A and Q1 for B, rises from 4 to 8. The returned assignment stays at cost 4 and becomes the unique minimum.

8. Change the overlap threshold

Take the two annotated boxes $(0, 0, 3, 3)$ and $(5, 0, 8, 3)$ as targets A and B. All boxes belong to the same class in one image. The prediction with score 0.90 is $(1, 0, 4, 3)$, a shifted version of A. Its shared rectangle with A is two units wide and three high. Each box has area 9, so their intersection over union (IoU) is

$$\text{IoU} = \frac{2 \cdot 3}{9 + 9 - 2 \cdot 3} = \frac{6}{12} = 0.50.$$

The 0.80 and 0.70 predictions are exact copies of A and B. The evaluator sorts predictions by descending score. For each prediction it checks the unused annotation with the highest IoU and accepts a match when the IoU is **at least** the threshold. Each annotation can be used once.

At 0.50 the first prediction matches A exactly at the threshold. The second prediction is a duplicate because A is already used. It has zero overlap with B. The third prediction matches B.

At 0.75 the first prediction falls below the threshold, so A is still available. The second prediction matches A, and the third prediction matches B.

After each prediction, precision divides the number of true positives so far by the number of predictions seen. Recall divides the same count by the two annotations. For example, by rank two each threshold has one true positive, so both fractions are $1/2$. The order in which the matches occur changes the curve.

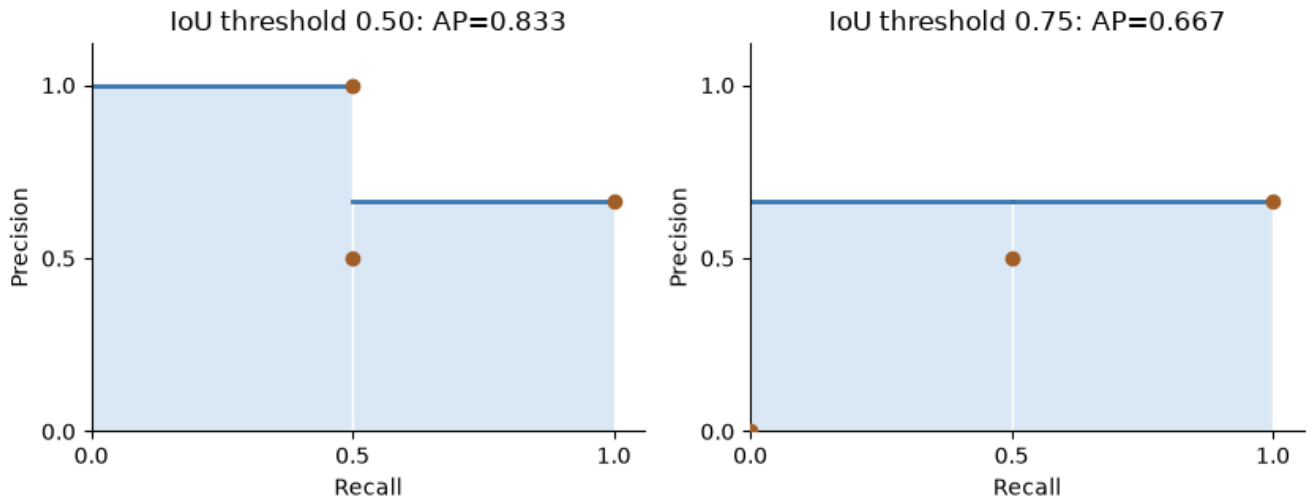
To compute all-points average precision (AP), take the best precision achieved at each recall level or higher. This gives a step function called the *precision envelope*. Its area is the sum of each recall increase times the envelope height on that interval. A rank that leaves recall unchanged adds zero width.

At 0.50 the envelope is 1 over the first half of recall and $2/3$ over the second half. At 0.75 the first prediction gives zero recall. At rank two the recall is $1/2$ and precision is $1/2$, but the later point at full recall improves precision to $2/3$. That later value sets the envelope height over both halves. Thus

$$\text{AP}_{0.50} = \tfrac{1}{2}(1) + \tfrac{1}{2}(\tfrac{2}{3}) = \tfrac{5}{6},$$
$$\text{AP}_{0.75} = \tfrac{1}{2}(\tfrac{2}{3}) + \tfrac{1}{2}(\tfrac{2}{3}) = \tfrac{2}{3}.$$

Both thresholds reach full recall. AP is lower by $1/6$ at 0.75 because the highest-scored prediction fails to match.

The score threshold and the overlap threshold answer different questions. The example varies the IoU required for a match while keeping all three predictions and their scores. Training's global minimum-cost assignment is not used here. The evaluator visits the ranked predictions sequentially. The plot shades the precision envelope for each IoU threshold, with raw points shown separately. Each shaded area spans recall from zero to one.



The left area is $5/6$ and the right area is $2/3$. At the stricter threshold, the later precision $2/3$ supplies the envelope over both recall increments. No trapezoidal interpolation is used. These are single-image, single-class, single-threshold all-points values, not COCO AP averaged over classes, IoU thresholds and its 101 recall samples.

9. Compare two prompting protocols (optional)

Use the same held-out images and reference instance masks in both protocols. Save image ID, instance ID, class and reference mask. For annotated-box prompts, use each instance's own annotated box. For detector-box prompts, freeze the detector and its confidence cutoff using validation data and generate boxes without consulting test annotations.

For evaluation, associate detector boxes to reference objects in descending confidence order within image and class. One explicit policy accepts the highest-IoU unused annotation when box IoU is at least 0.5. Record unmatched predictions and annotations separately. This evaluator uses reference labels to score predictions; it does not supply them to the detector.

Use the same segmenter, image preprocessing, coordinate conversion and selection rule in both protocols. If several masks are returned, choose the highest predicted mask-quality score with a fixed tie rule. Then measure its IoU with the reference mask on the original image grid. Selecting the highest measured IoU would use an oracle and answer a different question.

Report annotated-box mask IoU over all reference instances. Also compare both protocols on the same subset with an accepted detector box, pairing by instance ID. A mean difference on that subset isolates the prompt change among found objects. The full annotated-box protocol measures segmentation given the object's identity and location; it bypasses detection.

If there are two buses, the detector finds one, and its mask IoU is 0.8, the detected-only mean is 0.8. Coverage is $1/2$, and the all-instance mean with misses assigned zero is $(0.8 + 0)/2 = 0.4$. State the zero-for-miss rule explicitly. Extra predictions do not appear in that reference-instance mean, so report false positives separately, or supplement it with a complete instance-segmentation metric. Never silently drop missed objects from the end-to-end result. The fixture records these denominators; it does not run a detector or segmenter.

10. Audit resized annotations (optional)

Record original width and height, actual resized dimensions, and all padding amounts. Convert predictions from normalized or center-size form into corner coordinates on the padded canvas when needed. With scales s_x, s_y and left/top padding p_l, p_t, invert an edge using

$$x = (x' - p_l)/s_x, \qquad y = (y' - p_t)/s_y.$$

Apply the rule to both corners. Subtracting padding first matters: the offsets are measured on the enlarged canvas. Use actual dimension ratios if rounding changed the intended resize scale. Clip to original bounds if the scoring protocol requires it, then validate retained boxes. Avoid unnecessary integer rounding before measuring overlap.

Here the original image is 4×3 in width-by-height order. Resizing gives 8×6. With left padding 1 and top padding 2 and no right/bottom padding, the canvas is 9×8. Forward mapping multiplies by two and adds offsets: $(1, 0, 3, 2)$ becomes $(3, 2, 7, 6)$. Correct inversion recovers the original box and IoU 1. Division alone gives $(1.5, 1, 3.5, 3)$. Its intersection with the correct box has width 1.5 and height 1, hence area 1.5; each box has area 4. IoU is $1.5/(4 + 4 - 1.5) = 3/13$. The incorrect box is inside the original bounds, so a bounds-only check would miss the error.

For a mask, first map the output to the padded canvas according to the checkpoint's postprocessing. Remove the padding and resize the image region to the original grid. For discrete Boolean labels use nearest-neighbor resampling. For continuous logits or probabilities, follow the checkpoint's specified interpolation and foreground-threshold order.

As a tiny mask test, mark original pixel (row 1, column 2). Doubling each axis creates a 2-by-2 block. With the padding it occupies rows 4-5 and columns 5-6. Correct unpadding followed by nearest-neighbor downsampling recovers exactly the original grid. Taking a same-sized crop from the wrong origin can still return a 3-by-4 array while shifting its foreground pixel. The fixture uses integer repetition and stride sampling to make this exact nearest-neighbor example transparent; it is not a general image resizer.

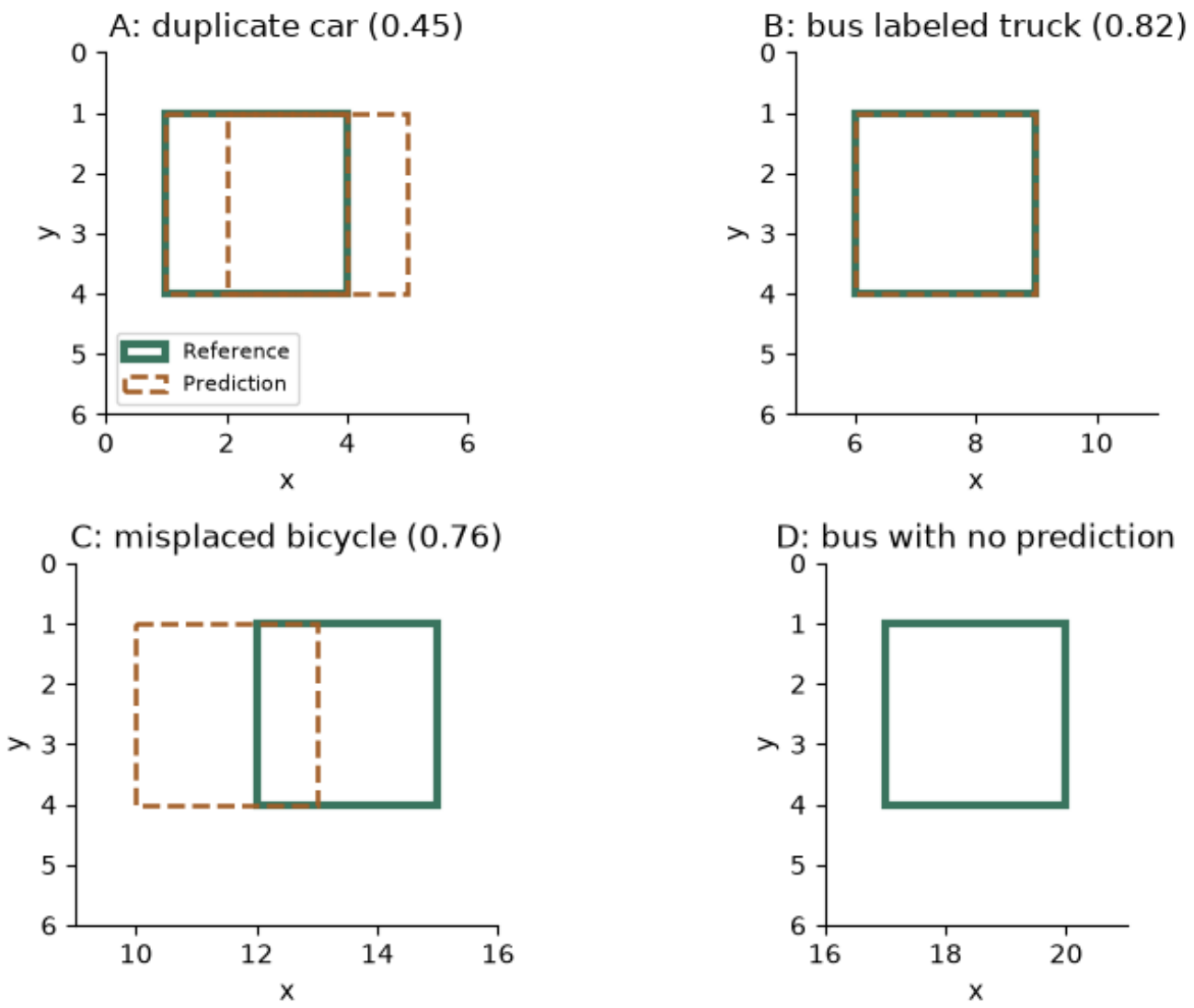
11. Separate detection errors (optional)

Use a 21×6 image containing a car A, bus B, bicycle C, and bus D. The four panels show the relevant portion of that same coordinate system. Solid green boxes are annotations; dashed orange boxes are predictions. Use IoU threshold 0.5 and confidence cutoff 0.4, with matching within class.

Object and reference box	Predictions	Error
A: car $(1, 1, 4, 4)$	Exact car at 0.95; car $(2, 1, 5, 4)$ at 0.45	Duplicate
B: bus $(6, 1, 9, 4)$	Exact box labeled truck at 0.82	Wrong class
C: bicycle $(12, 1, 15, 4)$	Bicycle $(10, 1, 13, 4)$ at 0.76	Misplaced box
D: bus $(17, 1, 20, 4)$	No candidate at any confidence	Missing instance

For A, the exact box claims the annotation first. The duplicate's IoU is $6/(9 + 9 - 6) = 0.5$, but A is already used. For B, IoU 1 cannot compensate for the wrong class: the truck prediction is a false positive and the bus remains unmatched. For C, IoU is $3/(9 + 9 - 3) = 0.2$, below the requirement; the prediction is false positive and the bicycle is missed. For D, no prediction exists to match.

Raising confidence from 0.4 to 0.5 removes the duplicate while keeping the correct car. It cannot change truck to bus, repair the bicycle's location, or invent D. A still-higher cutoff could hide the wrong predictions but would leave their objects missed. Lowering confidence can recover an instance only if a suitable candidate already exists below the cutoff; D has none. In other datasets, raising confidence can also remove correct low-scoring predictions. Inspect localization and class errors as well as the ranking.



Overlapping solid and dashed boundaries in B show that the coordinates are correct while the class is wrong. D has no dashed box. The four errors require different remedies; a single confidence threshold does not describe or repair all of them. The drawing is a labeled synthetic audit fixture.